Jay Baptista

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Education

- **Stanford University** — Stanford, CA
  Ph.D. Candidate in Physics and NSF Graduate Research Fellow — *Oct. 2023*
- **Yale University** — New Haven, CT
  Bachelor of Science - Astrophysics; GPA: 3.92 — *June 2019 - May 2023*

Research Experience

- **Graduate Student (Stanford University)** — Stanford, CA
  Wechsler Group — *Jan. 2024 - Present*
  Simulating globular cluster formation and evolution in the Symphony suite of cosmological simulations for Milky Way-mass galaxies.
- **Graduate Rotation Student (Stanford University)** — Stanford, CA
  Clark & Allen Groups — *Oct. 2023 - Jun. 2024*
  Characterized PAH emission in galaxies in PHANGS using Tensorial Minkowski Functionals; spectral fitting of AGN structural parameters using NuStar and XMM observations and forecasting reverberation mapping signals.
- **Undergraduate Research Fellow (Univ. of California Santa Cruz)** — Santa Cruz, CA
  Prochaska Group — *May 2022 - Aug 2022*
  Explored fast radio bursts as an independent measure of galactic feedback by using synthetic FRB surveys. We find that implementing galactic feedback as a free parameter causes loss of constraining power on the cosmological constant.
- **Undergraduate Research Fellow (University of Hawaii Mānoa)** — Mānoa, HI
  Sanderson Group — *May 2021 - Jul 2021*
  Comprehensively analyzed the evolution of dark matter halo symmetry axes in FIRE-2 galaxies over time and as a function of satellite galaxy interaction. Research is supervised and directed in collaboration with the Galaxy Dynamics lab at the University of Pennsylvania/Flatiron Institute's Center for Computational Astrophysics under Professor Robyn Sanderson.
- **Undergraduate Research Fellow (Yale University)** — New Haven, CT
  Geha Group — *May 2020 - July 2020*
  Evaluated the efficacy of stellar selection methods for determining star membership probabilities of ultra-faint Milky Way satellite galaxies using data from the Keck DEIMOS instrument. We find that color-magnitude and stellar position probabilities are insufficient in determining dispersion-based bound masses of ultra-faint galaxies.

Teaching Experience and Outreach

- **Teaching Assistant (Stanford University)** — Stanford, CA
  PHYSICS 100: Observational Astrophysics — *Spring 2026*
  Facilitated hands-on telescope operation at the Stanford Student Observatory; assessed astronomical imaging and data analysis assignments; mentored students through independent observational research projects.
- **Stanford Student Observatory** — Stanford, CA
  Public Observing Night Outreach — *2025 - Present*
  Engage the broader campus and local community in astronomy through regular public stargazing events at the Stanford Student Observatory.
- **Teaching Assistant (Stanford University)** — Stanford, CA
  PHYSICS 113: Computational Physics — *Spring 2025*
  Assessed problem sets and provided targeted feedback; led office hours on scientific computing; delivered a guest lecture on numerical methods for solving differential equations.
- **Teaching Assistant (Stanford University)** — Stanford, CA
  PHYSICS 16: Origin and Development of the Cosmos — *Spring 2024*
  Assessed homework and examinations; led study sections reinforcing foundational concepts in cosmology and astrophysics.
- **Peer Tutor** — New Haven, CT
  ASTR 310 — *February 2021 - May 2021*
  Taught a foundational understanding of galactic dynamics, chemistry, and substructure in one-on-one tutoring sessions. Additionally, I taught and reinforced theoretical concepts on black holes and active galactic nuclei.
- **Undergraduate Learning Assistant** — New Haven, CT
  ASTR 160 — *March 2022 - May 2022*
  Graded quizzes and provided feedback to students to assist in their understanding of introductory astronomy topics (i.e., orbital mechanics, rocket science, exoplanets, and black hole science). I also collaborated with instructors to develop a curriculum that helps build practical scientific intuitions for humanities and non-physical science majors.

Talks

- **American Astronomical Society (January 2026)** ***[Invited]***: "Beyond Gaia: Expanding the Dynamical Map of the Milky Way with Asteroseismic Distances." Splinter Session 247, Phoenix, AZ.
- **American Astronomical Society 53rd Division on Dynamics Astronomy (April 2022)**: Orientations of Dark Matter Symmetry Axes in Latte Galaxies
- **University of Hawaii Mānoa (July 2021)**: Orientations of Dark Matter Symmetry Axes in Latte Galaxies
- **Yale University (July 2020)**: Determining Membership Probabilities of Ultra-faint Dwarf Galaxies

Publications

- ***Measuring the Variance of the Macquart Relation in z-DM Modeling*** **(2023)**: **Baptista, J.**; Prochaska, J. X. ; Mannings, A. G. ; James, C. W.; Shannon, R. M.; Ryder, S. D.; Deller, A. T.; Scott, D. R.; Glowacki, M.; Tejos, N. Submitted to ApJ. arXiv:2305.07022.
- ***Orientations of DM Halos in FIRE-2 Milky Way-mass Galaxies*** **(2023)**: **Baptista, J.**; Sanderson, R.; Huber, D.; Wetzel, A; Sameie, O.; Chakrabarti, S.; Vargya, D.; Panithanpaisal, N.; Arora, A.; Cunningham, E. The Astrophysical Journal 958 44. DOI:10.3847/1538-4357/acea79.
- ***TOI-4010: A System of Three Large Short-Period Planets With a Massive Long-Period Companion*** **(2022)**: Kunimoto, M.; Vanderburg, A.; Huang, C.; et al. incl. **Baptista, J**. Submitted to ApJ.
- ***Constraining the Tilt of the Milky Way's Dark Matter Halo with the Sagittarius Stream*** **(2022)**: Panithanpaisal, N.; Sanderson, R.; Arora, A.; Cunningham, E.; **Baptista, J.** arXiv:2210.14983.

Skills Summary

- **Languages**: Python, C#, JavaScript, ADQL, Bash, Java

Observing Experience

- Keck I Telescope (1 night); PI: Daniel Huber
- Palomar Telescope (2 nights); PI: Marla Geha
- Keck II Telescope (3 nights); PI: J. X. Prochaska, Sunil Simha

Awards and Fellowships

- National Science Foundation Graduate Research Fellowship - October, 2023
- Stanford Graduate Fellowship - October, 2023
- Stanford Physics Department Fellowship - October, 2023
- Stanford Kavli Institute for Particle Physics and Astrophysics Fellowship - October, 2023
- University of California Santa Cruz: Lamat REU Fellow - June, 2022
- Yale Science Technology and Research Scholar II Fellow - August, 2021
- University of Hawaii Institute for Astronomy: REU Fellow - June 2021
- Yale Science Technology and Research Scholar Summer Fellow - May, 2020